Build Multimodal Generative AI Applications ×

Today's goal • 3/3 items

Module 1
Foundations of Multimodal AI

Module 2
Integrating Visual and Video Modalities

Module 3
Advanced Multimodal Applications

Build Advanced Multimodal Applications

- ✓ Introduction to Multimodal Retrieval-Augmented Generation (MM-RAG)
  Video • 6 min
- ✓ Lab: Build a Style Finder Using Multimodal Retrieval and Search
  Ungraded App Item • 45 min
- ✓ Multimodal Chatbots and QA Systems
  Video • 7 min
- ✓ Lab: Building Your First GenAI-Powered Image-Based Web Application: AI Nutrition Coach
  Ungraded App Item • 30 min
- ○ Build Advanced Multimodal Applications
  Practice Assignment • 15 min

Module Summary and Evaluation

- ○ **Summary and Highlights**
  Reading • 2 min
- ○ Cheat Sheet: Advanced Multimodal Applications
  Ungraded Plugin • 10 min
- ○ Graded Quiz: Advanced Multimodal Applications
  Graded Assignment • 21 min

Course Wrap-Up

- ○ Course Wrap-up
  Video • 3 min
- ○ Reading: Congratulations and Next Steps
  Reading • 2 min
- ○ Thanks from the Course Team
  Reading • 2 min

Summary and Highlights

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Congratulations! You have completed this module. At this point, you know that:

Multimodal Retrieval-Augmented Generation (MM-RAG)

- Combines multimodal inputs, such as text + images or videos, with retrieval-augmented generation, fetching relevant data to enhance LLM responses
- Pattern follows three steps:
  - Multimodal data retrieval
  - Contrastive learning for embeddings
  - Generative models informed by multimodal context
- Pipeline has four steps.
  - Data indexing: Diverse data, such as text, images, audio, and video, is converted into embeddings and stored in a vector database for efficient retrieval
  - Data retrieval: User query is embedded, and semantically relevant multimodal data is fetched from the vector database
  - Augmentation: Retrieved data is combined with the original query to enrich the context for generation
  - Response generation: Multimodal response is generated using the augmented input, blending information from all modalities

Multimodal Chatbots and QA Systems

- Advanced AI systems that process and respond to multiple data types, such as text, images, audio, and video.
- Can see, read, and understand the world more like humans do
- Key features:
  - Multiple input modalities
  - Integrated understanding
  - Contextual response generation
- Basic implementation steps:
  - Set up the environment and import libraries
  - Initialize the model
  - Prepare an image for processing.
  - Create the multimodal query function
  - Use the multimodal QA function

Mark as completed

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